

DETAILED DESCRIPTION OF THE INVENTION

Operational Workflow and Measurement Protocol

In various embodiments, the penile geometry profiling system operates according to a standardized measurement protocol configured to improve repeatability and reconstruction accuracy. Prior to scanning, the device may execute an initialization routine including sensor calibration, baseline distance verification, motion sensor zeroing, and optional environmental compensation adjustments. The system may prompt the user, via onboard indicators or a companion application, to position the device at a defined anatomical starting reference point.

The measurement sequence may be initiated manually or automatically once valid tissue presence and signal quality thresholds are detected. During scanning, the device captures synchronized radial distance measurements and longitudinal displacement data as the housing is moved along the anatomical axis in a controlled pathway. Sampling density may be dynamically adjusted based on detected motion speed, signal stability, or predefined resolution targets.

In certain embodiments, the system may require that scanning occur under predefined physiological conditions to promote consistency, such as a specified anatomical state, orientation, or support posture. The device may monitor motion continuity and signal confidence metrics to determine whether the scan meets minimum reconstruction criteria. If data integrity thresholds are not satisfied, the system may prompt the user to repeat the measurement sequence.

Throughout the scanning process, anti-spoofing verification logic may analyze physiological indicators, such as tissue compliance characteristics or vascular pulsatility patterns, to confirm measurement authenticity. Concurrently, encryption routines may secure collected data in memory prior to transmission or processing.

Upon completion of the scan pathway, the system aggregates the collected dataset and initiates the reconstruction engine to generate the three-dimensional geometric model and associated derived structural metrics. The finalized user-specific penile anatomical geometry profile may then be stored locally, transmitted securely to a companion device, or uploaded to a remote processing environment for additional analysis.

The operational workflow described herein may be modified to include multiple passes, segmented scanning regions, bidirectional motion paths, or adaptive recalibration steps without departing from the fundamental process of acquiring synchronized distance and motion data sufficient to generate a user-specific penile anatomical geometry profile.